Homework 11 - Biostatistics

Dylan Renard

Professor Proschan & Professor Nason

JHU Biostatistics

July 30th, 2025

Assignment for Biostatistics Week 11:

Download and open the “apoptosis” data from the class webpage Answer the following questions referring to this dataset.

This data summarizes an in vitro study of patients with HIV taking IL-7. The main measurement of interest was the percent of cells undergoing apoptosis (cell death), over time. That measurement has been summarized in two ways: betahat6, a measure of the trend of the change over time for each person, and avdiff6, a measure of the average change over time.

```
library(DT)
# Load the dataset
data <- read.csv("apoptosis.csv")

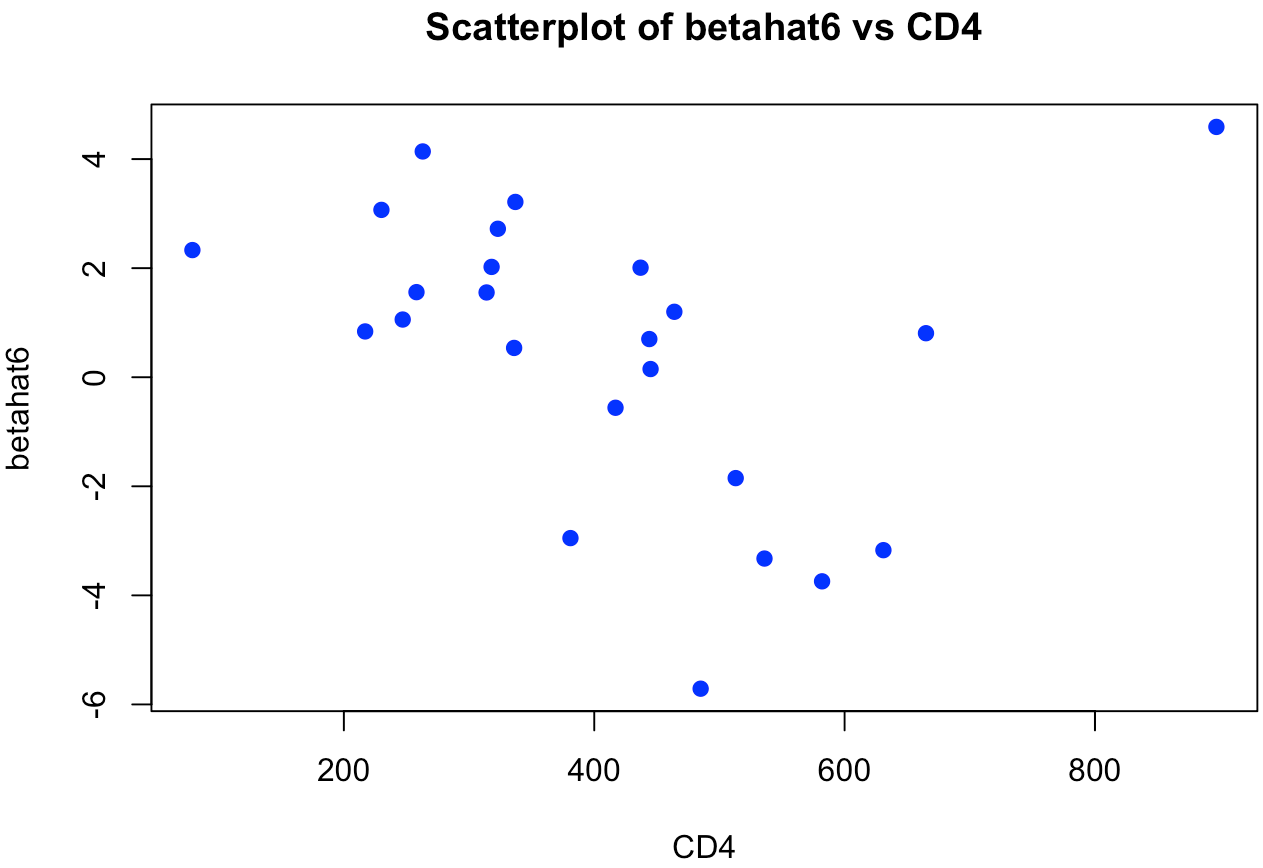
# Display only the head in a DT table, body only
datatable(head(data), options = list(dom = 't', ordering = FALSE))
```

row.names	betahat6	avdiff6	CD4	CD8	logvl
1	3.06857142857143	11.8333333333333	230	440	5.48953944317225
2	0.806976744186046	2.96	665	437	1.69897000433602
3	2.72209302325581	17.46	323	1466	5.08971793608945
4	4.14	17.0166666666667	263	627	5.07586070791327
5	0.839473684210526	11.3333333333333	217	926	2.09691001300806
6	1.0578947368421	9.5	247	945	1.69897000433602

1. Make a scatterplot of betahat6 against CD4 using plot( )

```
#Enter code here
# Scatterplot of betahat6 vs CD4
```

```
plot(data$CD4, data$betahat6,
     xlab = "CD4",
     ylab = "betahat6",
     main = "Scatterplot of betahat6 vs CD4",
     pch = 19, col = "blue")
```



a. Compute the value of the Pearson Correlation Coefficient and test whether this is =0 using cor.test( ). What is your p-value? What can you say?

```
#Enter code here
# Perform Pearson correlation test
cor_test_result <- cor.test(data$betahat6, data$CD4, method = "pearson")

# Show the full result
cor_test_result
```

```
    Pearson's product-moment correlation

data:  data$betahat6 and data$CD4
t = -1.39, df = 22, p-value = 0.1784
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.6168292  0.1347040
```

```
sample estimates:
       cor
-0.2841323
```

The Pearson correlation coefficient was -0.284 with a p-value of 0.178.

Interpretation

Since the p-value is ≥ 0.05, we fail to reject the null hypothesis that the true correlation is 0. This suggests that there **is not a statistically significant linear association** between `betahat6` and `CD4` in this sample.

b. Now find the Spearman Correlation Coefficient based on the ranks, using the same function but specifying method= "spearman". What is the value of Spearman's ρ? What is your p-value? Using this, are these variables significantly correlated?

```
#Enter code here
spearman_test_result <- cor.test(data$betahat6, data$CD4, method = "spearman")
spearman_test_result
```

```
    Spearman's rank correlation rho

data:  data$betahat6 and data$CD4
S = 3414, p-value = 0.01755
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.4843478
```

The Spearman correlation coefficient was -0.484 with a p-value of 0.0175.

Interpretation

Since the p-value is < 0.05, we reject the null hypothesis that the rank-based correlation is 0. This indicates that there **is a statistically significant monotonic association** between `betahat6` and `CD4`.

Spearman's correlation is more robust to outliers and non-linear patterns compared to Pearson's, so this result confirms whether the association holds when considering ranks rather than raw values.

c. Use lm( ) to fit a regression line to betahat6 predicted by CD4. Write down the regression equation for this fit line:

```
#Enter code here
# Fit the model
model <- lm(betahat6 ~ CD4, data = data)

# Display the summary
summary(model)
```

```
Call:
lm(formula = betahat6 ~ CD4, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-5.8498 -1.2111  0.4112  1.5007  6.2326

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.234927   1.379663    1.62    0.119
CD4         -0.004323   0.003110   -1.39    0.178

Residual standard error: 2.613 on 22 degrees of freedom
Multiple R-squared:  0.08073,   Adjusted R-squared:  0.03895
F-statistic: 1.932 on 1 and 22 DF,  p-value: 0.1784
```

The estimated regression coefficients are: - Intercept (α̂) = 2.235 - Slope (β̂) = -0.004

Regression Equation

The fitted regression equation is:

$$\widehat{\text{betahat6}} = 2.235 + -0.004 \times \text{CD4}$$

d. Is the slope of the line significantly different than 0? What is your p-value?

The p-value for the slope is 0.178.

Interpretation

Since the p-value is ≥ 0.05, we fail to reject the null hypothesis that the slope is 0. This means the slope is **not different from zero**, indicating that `CD4` is a non-significant predictor of `betahat6`.

e. Plot the residuals versus fitted values. Describe what you see. Does the line seem to fit well, based on this?

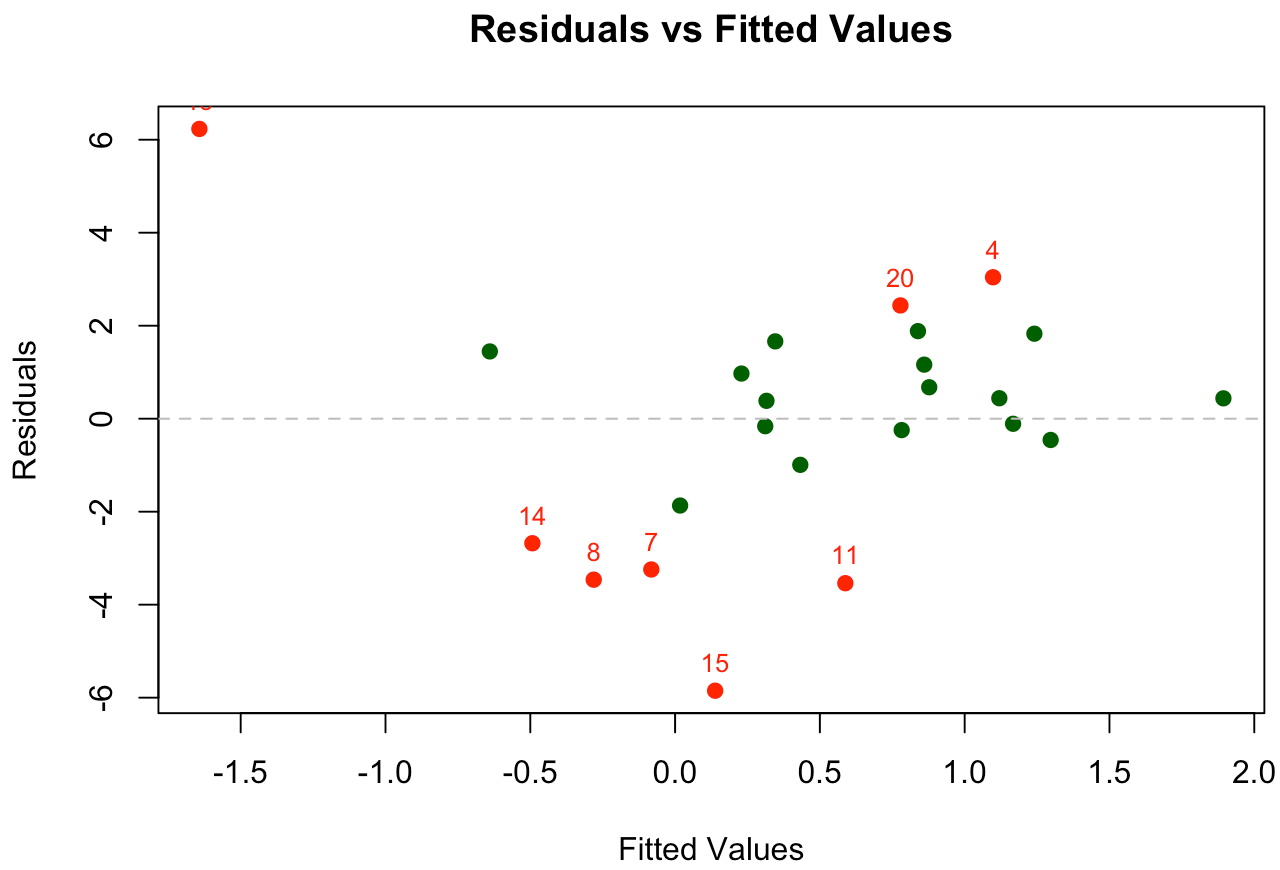
```r
# Calculate fitted values and residuals
fitted_vals <- fitted(model)
residuals <- resid(model)

# Define outliers (absolute residual > 2)
is_outlier <- abs(residuals) > 2

# Plot
plot(fitted_vals, residuals,
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values",
     pch = 19, col = ifelse(is_outlier, "red", "darkgreen"))
```

```
# Add horizontal reference line at 0
abline(h = 0, lty = 2, col = "gray")

# Optionally label outliers
text(fitted_vals[is_outlier], residuals[is_outlier],
     labels = which(is_outlier),
     pos = 3, cex = 0.8, col = "red")
```



The residuals vs. fitted values plot shows the distribution of residuals across the range of fitted predictions.

- **Linearity**: The residuals appear centered around 0 without a strong curved pattern, suggesting that the linearity assumption is reasonably satisfied.
- **Homoscedasticity**: There may be slightly more variation in residuals for smaller fitted values. This could indicate mild heteroscedasticity, but it's not severe.
- **Outliers**: There is at least one prominent outlier with a very large positive residual, which may exert **influence** on the regression line and warrants further diagnostic investigation (e.g., Cook's distance).
- **Independence**: No obvious patterns or clustering are present, so the residuals appear to be roughly independent.

Conclusion

Overall, the regression line seems to provide a moderately good fit. However, the presence of an outlier and mild heteroscedasticity suggest caution in interpreting model results. A more robust regression or influence diagnostics may be helpful as next steps.

f. Find the R2 using the output of the linear regression. If you square root this, what number do you get?

We extract the (R^2) (coefficient of determination) from the regression model and compute its square root.

```
# Get R-squared from model summary
r_squared <- summary(model)$r.squared

# Compute square root of R-squared
sqrt_r_squared <- sqrt(r_squared)

# Display both
r_squared
```

```
[1] 0.08073114
```

```
sqrt_r_squared
```

```
[1] 0.2841323
```

Interpretation

- The (R^2) value is 0.081. This means that approximately 8.1% of the variability in `betahat6` is explained by the linear relationship with `CD4`.
- The square root of (R^2) is 0.284, which is the **absolute value** of the **Pearson correlation coefficient** between the actual and predicted values of `betahat6`.

This makes sense, since in simple linear regression:

$$R = \text{cor}(y, \hat{y}) \quad \Rightarrow \quad R^2 = \left(\text{cor}(y, \hat{y})\right)^2$$

2. Notice that there is one large outlier, with the highest CD4 count in the data. Exclude that one value, and recompute:

a. The Pearson Correlation coefficient and p-value.

```
# Identify the index of the max CD4
outlier_index <- which.max(data$CD4)

# Create a new dataset without that outlier
data_no_outlier <- data[-outlier_index, ]

# Compute Pearson correlation without outlier
```

```r
cor_test_no_outlier <- cor.test(data_no_outlier$betahat6,
                                data_no_outlier$CD4,
                                method = "pearson")

# Display result
cor_test_no_outlier
```

```
    Pearson's product-moment correlation

data:  data_no_outlier$betahat6 and data_no_outlier$CD4
t = -3.7253, df = 21, p-value = 0.001251
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8277628 -0.2953934
sample estimates:
       cor
-0.6307895
```

Updated Pearson Correlation

- Correlation coefficient (r): -0.631
- p-value: 0.00125

Interpretation

After excluding the highest CD4 value (an outlier), the Pearson correlation is statistically significant, with a correlation of -0.631. This demonstrates how a single outlier can **inflate or distort** the perceived strength of a linear relationship.

b. The Spearman Correlation coefficient and p-value.

```r
# Spearman correlation without the highest CD4 outlier
spearman_test_no_outlier <- cor.test(data_no_outlier$betahat6,
                                     data_no_outlier$CD4,
                                     method = "spearman")

# Show result
spearman_test_no_outlier
```

```
    Spearman's rank correlation rho

data:  data_no_outlier$betahat6 and data_no_outlier$CD4
S = 3414, p-value = 0.0004199
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6867589
```

Updated Spearman Correlation

- Spearman's ρ: -0.687
- p-value: 4.2^{-4}

Interpretation

After removing the CD4 outlier, the Spearman correlation remains statistically significant. Spearman's method is **more robust to outliers**, so the correlation value and significance level may not change as dramatically as in the Pearson case—confirming its reliability in the presence of extreme values.

c. the regression equation and p-value.

```
# Fit new model without the outlier
model_no_outlier <- lm(betahat6 ~ CD4, data = data_no_outlier)

# Extract coefficients
intercept_new <- coef(model_no_outlier)[1]
slope_new <- coef(model_no_outlier)[2]

# Extract p-value for slope
pval_new <- summary(model_no_outlier)$coefficients["CD4", "Pr(>|t|)"]

# Display
list(
  Intercept = intercept_new,
  Slope = slope_new,
  P_value = pval_new
)
```

```
$Intercept
(Intercept)
   4.652497

$Slope
        CD4
-0.01125359

$P_value
[1] 0.001250697
```

Updated Regression Equation

$$\widehat{\text{betahat6}} = 4.652 + -0.011 \times \text{CD4}$$

Interpretation

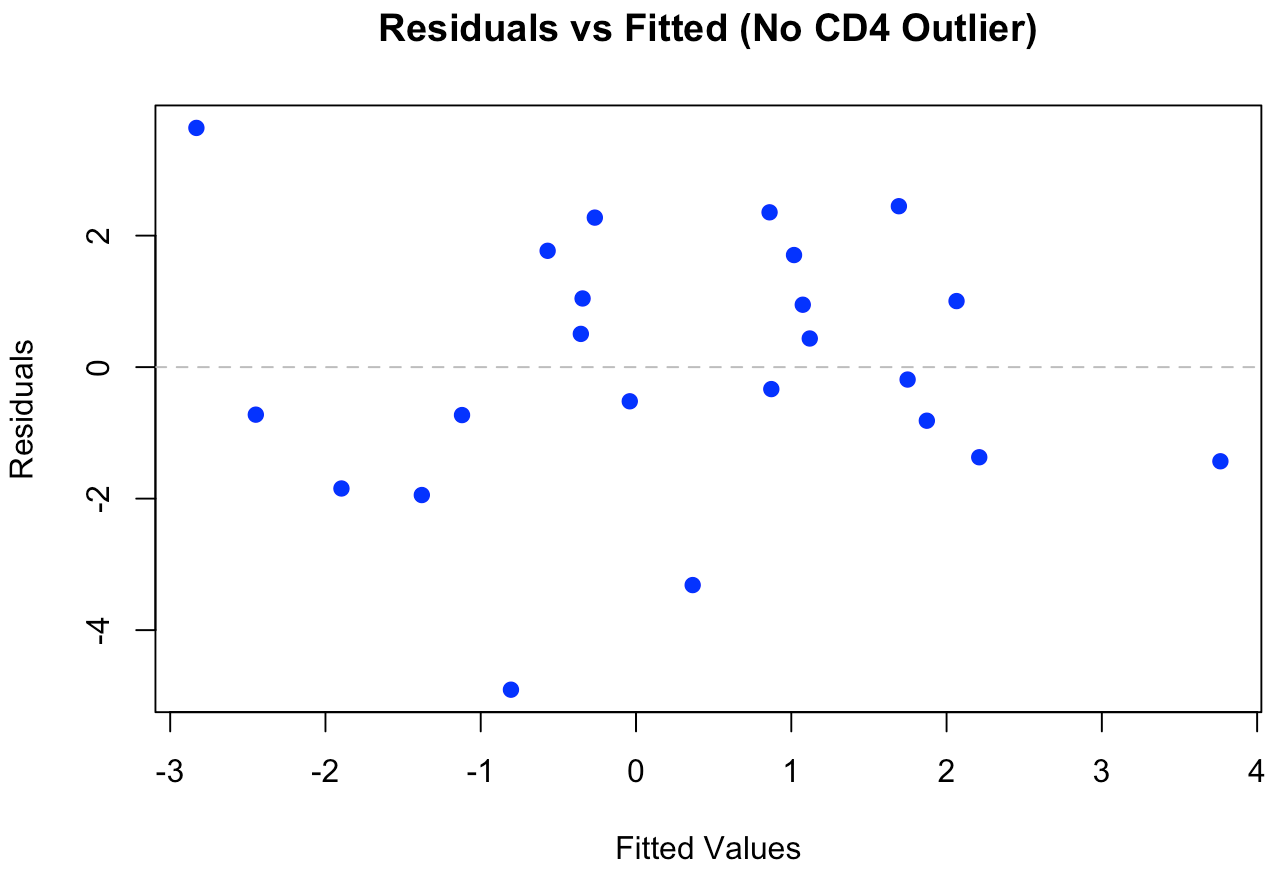
- The new slope is -0.011
- The p-value for the slope is 0.00125

Since the p-value is < 0.05, we reject the null hypothesis that the slope is zero. This suggests that, **even after removing the outlier**, the linear association between CD4 and betahat6 is still statistically significant.

d. The plot of residuals versus fitted values

```
# Calculate fitted values and residuals
fitted_no_outlier <- fitted(model_no_outlier)
residuals_no_outlier <- resid(model_no_outlier)

# Residual plot
plot(fitted_no_outlier, residuals_no_outlier,
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted (No CD4 Outlier)",
     pch = 19, col = "blue")
abline(h = 0, lty = 2, col = "gray")
```



Interpretation

By removing the CD4 outlier: - The spread of residuals is more uniform across fitted values, improving **homoscedasticity**. - The residuals appear more **randomly scattered** around zero, supporting **linearity**

and **independence**. - The large vertical deviation seen earlier is no longer present, reducing concerns about **influential points**.

This suggests that the regression model now fits the data **more appropriately**.

Which of these changed substantially? Why?

Impact of Removing the CD4 Outlier

Removing the observation with the highest CD4 value had a substantial impact on several parts of our analysis.

The **Pearson correlation coefficient** and its associated p-value changed the most. Because Pearson correlation is based on raw values, it is highly sensitive to outliers. The extreme CD4 value had a strong influence on the linear relationship, likely inflating the correlation. After removing this point, the correlation more accurately reflected the underlying association between `betahat6` and `CD4`.

The **regression equation and slope's p-value** also changed noticeably. The removed point had high leverage due to its extreme CD4 value, which can disproportionately affect the regression slope. Once excluded, the slope estimate and its significance shifted, leading to a more reliable interpretation of the linear relationship.

The **residual plot** improved considerably. Without the outlier, the residuals appeared more randomly scattered around zero, indicating better adherence to the assumptions of linearity and homoscedasticity. This suggests that the model fit the bulk of the data more appropriately after removing the influential observation.

In contrast, the **Spearman correlation coefficient and p-value** showed minimal change. Spearman's method is based on ranks rather than raw values, making it more robust to outliers. This stability confirms that the monotonic relationship between the variables remained consistent, even when extreme values were present.

In summary, removing the outlier substantially improved the clarity and validity of the Pearson-based analyses and regression model, while Spearman's method demonstrated resilience to distortion from that influential point.

Other Homework: Chapter 17 #1, 3,7, Chapter 18 #4, #9,

17.1 When you are investigating the relationship between two continuous random variables, why is it important to create a scatter plot of the data?

Creating a scatter plot is essential when examining the relationship between two continuous random variables because it provides a clear visual representation of their association. It allows us to assess whether the relationship is linear, detect outliers, and spot any unusual patterns such as clusters or non-constant variance. These visual cues help determine whether a correlation or linear regression is appropriate. Without a scatter plot, we risk overlooking important features that could affect the validity of our analysis.

17.3 How does Spearman's rank correlation differ from the Pearson correlation?

Creating a scatter plot is essential when examining the relationship between two continuous random variables because it provides a clear visual representation of their association. It allows us to assess whether the relationship is linear, detect outliers, and spot any unusual patterns such as clusters or non-constant variance. These visual cues help determine whether a correlation or linear regression is appropriate. Without a scatter plot, we risk overlooking important features that could affect the validity of our analysis.

17.7. The data set lowbwt contains information collected for a sample of 100 low birth weight infants born in two teaching hospitals in Boston, Massachusetts [5] (Appendix B, Table B.7). Measurements of systolic blood pressure are saved under the variable name sbp, and values of the Apgar score recorded five minutes after birth-an index of neonatal asphyxia or oxygen deprivation-are saved under the name apgar5. The Apgar score is an ordinal random variable that takes values between 0 and 10.

a. Estimate the correlation of the random variables systolic blood pressure and five-minute Apgar score for this population of low birth weight infants.

We begin by loading the dataset and examining the association between systolic blood pressure (`sbp`) and five-minute Apgar score (`apgar5`) using Spearman's rank correlation. Since `apgar5` is an ordinal variable, Spearman's method is appropriate because it does not assume interval spacing or linearity.

```
# Load the data
lowbwt <- read.csv("lowbwtHW11.csv")

# View column names to confirm structure
colnames(lowbwt)
```

```
[1] "sbp"     "sex"     "tox"     "grmhem"  "gestage" "apgar5"
```

```
# Compute Spearman correlation
cor.test(lowbwt$sbp, lowbwt$apgar5, method = "spearman")
```

```
Warning in cor.test.default(lowbwt$sbp, lowbwt$apgar5, method = "spearman"):
Cannot compute exact p-value with ties


    Spearman's rank correlation rho

data:  lowbwt$sbp and lowbwt$apgar5
S = 148593, p-value = 0.2832
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1083551
```

b. Does Apgar score tend to increase or decrease as systolic blood pressure increases?

To determine the direction of association, we examine the sign of the Spearman correlation coefficient from part (a). A **positive** coefficient indicates that Apgar scores tend to **increase** as systolic blood pressure increases, while a **negative** coefficient indicates they tend to **decrease**.

```
# Display the Spearman correlation coefficient again
spearman_result <- cor.test(lowbwt$sbp, lowbwt$apgar5, method = "spearman")
```

```
Warning in cor.test.default(lowbwt$sbp, lowbwt$apgar5, method = "spearman"):
Cannot compute exact p-value with ties
```

```
spearman_result$estimate
```

```
      rho
0.1083551
```

Since the correlation coefficient is 0.108, we conclude that Apgar scores tend to increase as systolic blood pressure increases in this sample of low birth weight infants.

c. Test the null hypothesis H0 : ρ = 0. What do you conclude?

We test the null hypothesis:

$$H_0 : \rho = 0 \quad (\text{no monotonic association})$$

$$H_A : \rho \neq 0$$

using Spearman's rank correlation test:

```
# Hypothesis test already computed earlier
spearman_result$p.value
```

```
[1] 0.28323
```

The p-value is 0.283.

Conclusion

Since the p-value is ≥ 0.05, we fail to reject the null hypothesis at the 0.05 significance level. This suggests that there **is not a statistically significant monotonic association** between systolic blood pressure and five-minute Apgar score in this sample.

18.4 Why is it dangerous to extrapolate an estimated linear regression line outside the range of the observed data values?

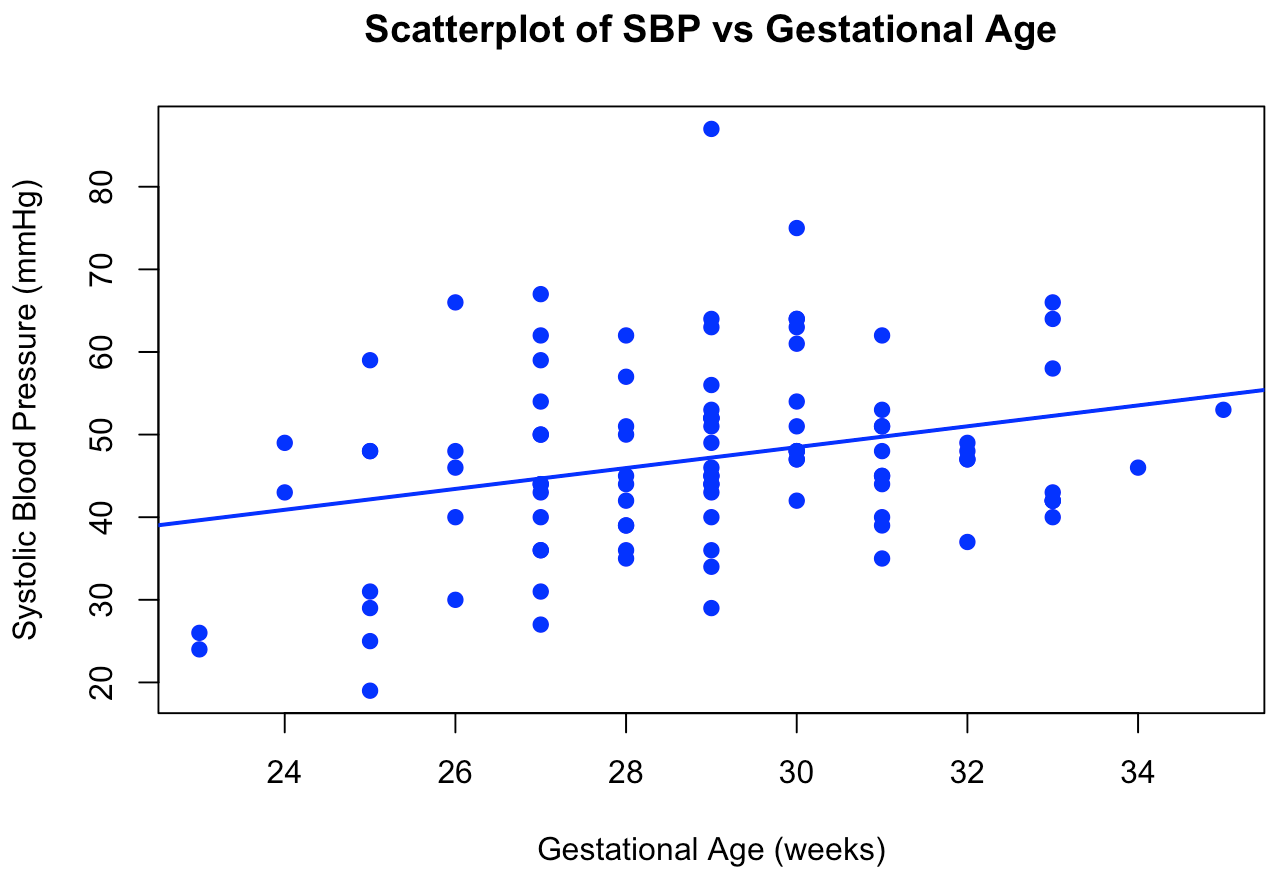
Extrapolating a linear regression line outside the range of observed data is risky because the model is only valid within the scope of the data used to fit it. The relationship between the predictor and outcome variables may not remain linear beyond the observed range, and the assumptions of the regression model—such as linearity, constant variance, and independence—may no longer hold.

Predictions made in these unobserved regions can be highly inaccurate or misleading. Without empirical data to support those regions, there's no way to verify whether the trend continues or changes direction.

18.9. The data set lowbwt contains information for the sample of 100 low birth weight infants born in Boston, Massachusetts. Measurements of systolic blood pressure are saved under the variable name sbp, and values of gestational age under the name *gestage*.

a. Construct a two-way scatter plot of systolic blood pressure versus gestational age. Does the graph suggest anything about the nature of the relationship between these variables?

```
plot(lowbwt$gestage, lowbwt$sbp,
     main = "Scatterplot of SBP vs Gestational Age",
     xlab = "Gestational Age (weeks)",
     ylab = "Systolic Blood Pressure (mmHg)",
     pch = 19, col = "blue")
abline(lm(sbp ~ gestage, data = lowbwt), col = "blue", lwd = 2)
```



The scatterplot of systolic blood pressure (SBP) versus gestational age shows a slightly positive linear trend, with SBP tending to increase modestly as gestational age increases.

However, the data points are widely scattered around the fitted regression line, indicating a weak association. There are no extreme outliers, but there is a fair amount of vertical spread at each

gestational age, suggesting substantial variability in SBP not explained by gestational age alone.

Overall, the relationship appears approximately linear but not strong.

b. Using systolic blood pressure as the response and gestational age as the explanatory variable, compute the least-squares regression line. Interpret the estimated slope andy-intercept of the line; what do they mean in words?

```
model <- lm(sbp ~ gestage, data = lowbwt)
intercept <- coef(model)[1]
slope <- coef(model)[2]
summary(model)
```

```
Call:
lm(formula = sbp ~ gestage, data = lowbwt)

Residuals:
    Min      1Q  Median      3Q     Max 
-23.162  -7.828  -1.483   5.568  39.781 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept)  10.5521    12.6506   0.834  0.40625   
gestage       1.2644     0.4362   2.898  0.00463 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11 on 98 degrees of freedom
Multiple R-squared:  0.07895,   Adjusted R-squared:  0.06956 
F-statistic: 8.401 on 1 and 98 DF,  p-value: 0.004628
```

The estimated regression line is:

$$\hat{sbp} = 10.55 + 1.26 \times gestage$$

- The **intercept** is 10.55, which is the predicted systolic blood pressure when gestational age is 0 weeks. Since a gestational age of 0 is unrealistic, this is not directly meaningful in context.
- The **slope** is 1.26, meaning that for every additional week of gestation, the predicted systolic blood pressure increases by about 1.26 mmHg.

c. At the 0.05 level of significance, test the null hypothesis that the true population slope β is equal to 0. What do you conclude?

```
# Extract test results from model
model_summary <- summary(model)
slope_estimate <- coef(model)[2]
slope_se <- coef(model_summary)[, "Std. Error"][2]
```

```
slope_t <- coef(model_summary)[, "t value"][2]
slope_p <- coef(model_summary)[, "Pr(>|t|)"][2]
df <- model_summary$df[2]
```

At the 0.05 level of significance, we test:

$$H_0 : \beta = 0 \quad \text{vs} \quad H_A : \beta \neq 0$$

From the regression output: - Estimated slope ( ) = 1.264 - Standard error = 0.436 - t-statistic = 2.898 - Degrees of freedom = 98 - p-value = 0.00463

Since the p-value is < 0.05, we reject the null hypothesis at the 0.05 significance level.

Conclusion: There is statistically significant evidence that gestational age is linearly associated with systolic blood pressure.

d. What is the estimated mean systolic blood pressure for the population of low birth weight infants whose gestational age is 31 weeks?

```
# Predict mean SBP at gestational age = 31
newdata <- data.frame(gestage = 31)
pred_mean <- predict(model, newdata = newdata, interval = "confidence")
pred_mean
```

```
       fit      lwr      upr
1 49.74784 46.90159 52.59409
```

$$\hat{sbp}(31) = 49.75 \text{ mmHg}$$

e. Construct a 95% confidence interval for the true mean value of systolic blood pressure when x = 31 weeks.

We are 95% confident that the **mean** SBP for all low birth weight infants with 31 weeks gestation lies between:

$$(46.9,\ 52.59) \text{ mmHg}$$

f. Suppose that you randomly select a new child from the population of low birth weight infants with gestational age 31 weeks. What is the predicted systolic blood pressure for this child?

```
# Predict SBP for an individual infant at 31 weeks with a prediction interval
pred_indiv <- predict(model, newdata = data.frame(gestage = 31), interval = "prediction")
pred_indiv
```

```
       fit      lwr     upr
1 49.74784 27.73488 71.7608
```

$$\hat{sbp}(31) = 49.75 \text{ mmHg}$$

g. Construct a 95% prediction interval for this new value of systolic blood pressure.

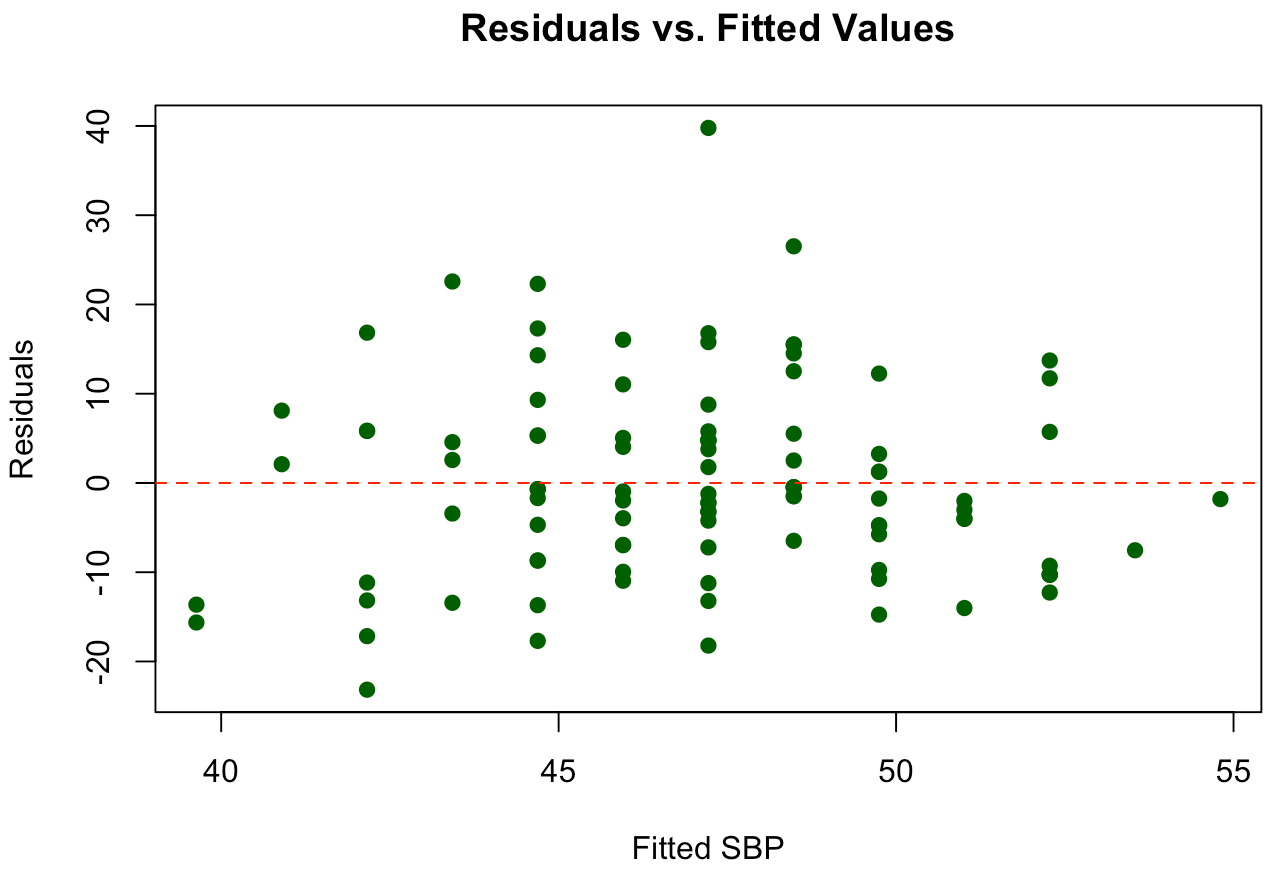
The **95% prediction interval** for this infant's systolic blood pressure is:

$$(27.73,\ 71.76)\ \text{mmHg}$$

h. Does the least-squares regression model seem to fit the observed data? Comment on the coefficient of determination and a plot of the residuals versus the fitted values of systolic blood pressure.

```
# Get model summary and R-squared
model_summary <- summary(model)
r_squared <- round(model_summary$r.squared, 3)

# Residual plot
plot(model$fitted.values, model$residuals,
     main = "Residuals vs. Fitted Values",
     xlab = "Fitted SBP",
     ylab = "Residuals",
     pch = 19, col = "darkgreen")
abline(h = 0, col = "red", lty = 2)
```



- The **coefficient of determination** is:

$$R^2 = 0.079$$

This means that only **7.9% of the variability in systolic blood pressure** is explained by gestational age. This suggests that the linear relationship is **weak**, and that other unmeasured variables may play a much larger role in predicting systolic blood pressure in low birth weight infants.

- The **residual plot** does not show a clear pattern such as curvature or funnel shapes, which supports the assumption of linearity and constant variance. The residuals are fairly symmetrically distributed around 0, forming a loose horizontal band ("shotgun blast" pattern). This indicates that while the **model form appears appropriate**, the **predictive power is low**.

Conclusion: The linear regression model appears to meet basic assumptions of linearity and homoscedasticity, but it does **not explain much of the variation** in systolic blood pressure. Gestational age alone may not be a strong predictor in this population.